AI-Based Smart Civic Issue Detection and Complaint System

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***Abstract*—Rapid urbanization has increased the number of civic issues such as potholes, waste accumulation, waterlogging, and damaged street infrastructure. Traditional complaint systems rely heavily on manual reporting, leading to inconsistent data, delayed responses, and poor issue prioritization. This paper presents CivicAI, an AI-powered civic help system designed to automate and streamline the complaint reporting process. It creates a detailed complaint message, captures live GPS coordinates, and provides a complaint ID for tracking. The backend features a robust data management system that utilizes Node.js and Express.js, and the frontend uses HTML, CSS, and JavaScript to ensure the interface is user-friendly and seamless. Further, we set up an Admin Dashboard to monitor complaints and the status of their resolutions. On testing, it was found that image classification was correct, data capture reliable, and response times fast, making this solution perfect for real-world civic problem-solving. This project delivers a framework that is scalable, privacy-conscious, and user-friendly to bridge the gap between citizens and the local authority, thus increasing the responsiveness of civic services by using smart automation..**

Keywords: Artificial Intelligence (AI), Gemini 2.5 Flash Model, Civic Complaint Management, Image classification, Natural Language Processing(NLP)

I. Introduction

Urbanization has caused a rapid increase in the number of civic problems that local authorities face, such as potholes, waste accumulation, waterlogging, and broken streetlights. [1], [2] Ideally, to ensure that the infrastructure of the city does not deteriorate and citizen satisfaction is maintained, there is a dire need for efficient reporting of civic problems and their resolution. However, all existing grievance redressal systems are mainly manual, irregular, and cumbersome. [4] Citizens must report a problem through detailed forms or calls, which usually end up in incomplete or incorrect submission on their part. The lack of standardized complaint categories and inconsistency in location details further worsen the situation, thus delaying the assignment and resolution of the issues. As a result, the gap between citizens and municipal authorities will continue to widen and affect the overall efficiency of civic maintenance.

The rapid evolution of artificial intelligence (AI) and computer vision provides an increasing opportunity for better identification, categorization, and management of civic issues. AI-driven automation will bring more accuracy, reduce human dependency, and guaranty response times. Image recognition models directly identify the type of civic issue in the photograph.

The proposed system, CivicAI, addresses these challenges by providing a platform powered by AI to automatically manage civic complaints. The system captures the issue by taking an image and uploading it. Image processing uses Google Gemini Vision AI to classify complaints based on predefined categories such as potholes, garbage, or lighting failure. It also includes the creation of a short complaint description, captures live GPS coordinates, and assigns a unique complaint ID for tracking. The application backend was set up using Node.js and Express.js to ensure smooth data management, while the frontend used HTML, CSS, and JavaScript offers a simple, responsive interface. A dedicated administrative dashboard allows authorized officials to efficiently review, update, and close grievances.

This paper presents the design and implementation of CivicAI, focusing on how the system bridges the gap between citizens and municipal authorities through intelligent automation. The work intends to exemplify the role of AI in smart governance by enhancing transparency, accuracy, and timeliness regarding civic issue management.

II. Importance of technology

Technology is playing a huge role in upgrading the traditional civic management system to be more efficient, transparent, and data-driven. Considering manual complaint handling processes related to urban governance, human errors, response delay, and improper coordination among various departments act as hurdles to the concerned processes. [4] Artificial Intelligence and new Web technologies alleviate these types of bottlenecks by automating tasks that are mundane, thus ensuring seamless information exchange between citizens and authorities. Powered by advances in AI, especially computer vision and natural language processing, systems can identify and describe real-world issues from either images or textual input with minimum human intervention. On the other hand, leveraging those capabilities, civic systems can classify complaints, generate structured descriptions, and provide actionable insight to decision-makers. Similarly, technologies such as GPS-based geo-location enhance the accuracy of

issue reporting, while Cloud-based frameworks ensure real-time access and scalability. [9] This will not only reduce manual effort but also introduce greater accountability and transparency in civic operations. For the citizens, this would mean ease of reporting; for the authorities, a unified, intelligent platform to monitor and address these issues. [5], [12]

III. Literature review

Kormpho et al. [1] developed a "Smart Complaint Management System", which digitized grievance handling using a web-based interface. Their work proposed a structured platform for citizens to report issues like sanitation, road damage, and streetlight faults. While the system enhanced access and transparency, it still maintained much manual classification and administrative verification.

Karukayil et al. [2] presented a "deep learning-based pothole detection model" that integrates Vision with LiDAR data. Their results showed that the feature extraction via multimodal improves the accuracy considerably and enhances real-time road surface analysis for quicker maintenance decisions.

Yang et al. [3] proposed a "garbage classification system using the YOLOv5 model" that employed object detection for the sorting of various categories of waste. The paper reports very high precision under different environmental conditions, which proved the feasibility of a CNN for civic applications.

Batty et al. [4] reviewed the general concept of smart cities driven by analytics with integrated technologies. This paper noted how combining AI with real-time data collection creates greater efficiency and visibility in civic governance.

Russell and Norvig [5] gave the fundamental theories in their milestone textbook entitled "Artificial Intelligence: A Modern Approach", which includes reasoning, knowledge representation, and decision-making. Their concepts theoretically underpin the applications that have been done using AI-driven automation and help shape modern implementations in civic complaint management and public service optimization.

Abadi et al. [6] produced TensorFlow, a large-scale open-source framework supporting machine learning applications such as image recognition, anomaly detection, and predictive analytics. The architecture of TensorFlow allows for rapid prototyping of AI models, including those used in CivicAI.

Sun et al. [7], on extending this, proposed an improved "garbage detection architecture, namely YOLOv5-OCDS", aimed at optimizing feature extraction as well as improving the classification speed. It improved the rates of detection with less computational complexity and hence is fit for use in low-power embedded devices in the smart city infrastructure.

Zhou et al. [8], where "YOLOv8" was employed with anchor-free detection mechanisms to enhance recognition accuracy even for complex backgrounds.

Rakib and Shamsuzzaman [9]. propose "YOLObin" for the classification of non-decomposable waste using YOLOv7. Their main observation from the experiment was that the robustness in waste segregation performance depends on dataset diversity and hyperparameter optimization.

S. V. J. et al. [10] proposed a multi-object detection model fusing Region Proposal Networks with YOLO for effective solid waste classification. Indeed, the approach improved the detection accuracy and speed, hence setting a solid foundation for modern AI-driven waste management systems such as CivicAI in 2022 as published in Global NEST Journal.

Bidve et al. [11] proposed a "computer vision-based pothole detection system" that employed machine learning for the identification of road damages from real-time camera feeds. The work reached a very high detection precision with low false alarm rates, thus providing the authorities with an efficient way to improve road safety and allocate resources effectively in repair activities.

Musale et al. [12] have proposed a Plastic Waste Detection System using the YOLOv5 deep learning object detection algorithm, which identifies different types of plastic waste among complex backgrounds. This work has been highlighted in LNCS, in which more consideration is given to the ever-increasing importance of AI in waste management towards sustainability.

Musale and Kadam [13] developed UIM-DUDnCNN, a hybrid deep learning approach where illumination map-based denoising is combined with an InceptionV3-based transfer learning model for face recognition. Their work has been published in The Imaging Science Journal, proving that the integration of illumination normalization with deep convolutional neural networks can improve image clarity significantly and enhance the preciseness of recognitions under varying light conditions.

Musale and Kadam [14] proposed the ILMNetCNN, an optimization-enabled incremental learning-based MobileNet framework for face detection and recognition from drone imagery. Incremental learning in their model adaptively improved the accuracy without retraining from scratch, offering a highly efficient approach for processing large-scale visual data.

IV. Advantage of proposed model over existing model

The CivicAI model represents a number of advantages in the building, performance, accuracy, and usability that distinguish civic complaint management systems. First, the mentioned system provides integration of Artificial Intelligence into the process of complaint handling, starting from issue identification to report generation. Traditional systems mostly rely on manual inputs, which cause more delays and result in inconsistent reports. Whereas CivicAI employs AI-based image classification that would automatically detect the type of civic issues-pothon roads, garbage accumulation, waterlogging, streetlight damage among others. Such automation cuts human dependency, reduces erroneous complaints, and categorizes complaints much faster.

Another great advantage of this system is its real-time geolocation capture: integrating GPS-based location tracking can link every complaint to an accurate geographical position. [7], [10] This helps the municipal corporations to accurately point out the areas where such a problem was reported, which

helps reduce the time required to respond to and resolve the problem. The timestamp and automatic ID also helps enhance traceability, which ensures that no duplicate entries occur and making it easier to keep track of every complaint within the system.

The interface of CivicAI is intuitive and user-friendly for all its users. Using this application, citizens can report any complaint just by uploading a single photo-a tap away-since there is no need to fill out long forms or give detailed textual information. This simplicity will encourage better public participation and breed a significant rise in civic efficiency even among people who aren't exactly confident with technology. Its front end has been developed using HTML, CSS, and JavaScript, and it assures responsiveness on anything from smartphones to desktops.

The architecture of the backend was developed with Node.js and Express.js, ensuring it was lightweight and fully scalable. This will allow the client interface to smoothly communicate with both the AI module and the storage components with ease. Its modular design allows for future upgrades, such as advanced database integrations, multilingual interfaces, or predictive analytics tools, with minimum fuss. In using JSON-based storage for the prototype, CivicAI demonstrates how low-cost, easily deployable systems can function without relying on complex infrastructure. [6], [11]

Another important benefit involves AI-powered text generation that automatically generates structured and meaningful complaint descriptions, depending on the classified issue. Uniformity in every submission is ensured with no room for ambiguity due to manually written complaints. Secondly, the admin dashboard provides centralized control to monitor, manage, and resolve issues. Authorities can filter complaints according to category, check their status, assign tasks, and mark completion for better transparency and accountability.

From a broader perspective, CivicAI also contributes to smart governance by promoting data-driven decision-making. The system's ability to store, categorize, and visualize complaints allows city administrators to identify recurring problems, high-risk zones, and maintenance patterns. This data can be leveraged for long-term urban planning and resource optimization. In all, CivicAI embodies the union of automation, accuracy, and accessibility in effectively offering an integrated, scalable, user-centric civic management solution. It help bridge communications gaps between citizens and authorities, helps enhance the efficiency of operations, and supports the development of the sustainable technology-driven smart cities. [8], [9], [14]

V. ARCHITECHTURE DIAGRAM

The system architecture of the proposed AI-powered Civic Help System, CivicAI, is organized into four main layers: Presentation Layer, Application Layer, Data Layer, and External Services. Each layer has different functions that, together, ensure smooth and automated operation. [6], [8]

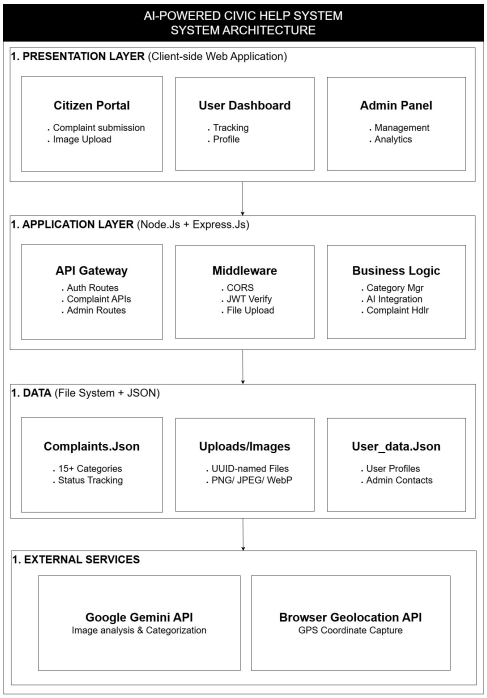


Fig. 1. Architecture of Proposed AI-Powered Civic Help System

Presentation Layer: This layer is the web interface that the users will directly face, including three key modules: the Citizen Portal, User Dashboard, and Admin Panel. It provides the means for citizens to submit complaints with images, track their reports, and administrators to manage and analyze system data. The Application Layer in Node.js and Express.js acts as the core processing unit, housing the API Gateway for routing and authentication, Middleware for handling cross-origin requests, file uploads, and token verification, and Business Logic for complaint category management, integrating AI, and problem processing efficiently.

The Data Layer maintains system information in structured JSONs, holding data related to complaint records, user data, and uploaded images with unique identifiers. Finally, the External Services Layer integrates Google Gemini API for image classification and Browser Geolocation API for GPS-based location capture. [3], [7]

This layered architecture provides scalability, modularity, and high accuracy, allowing CivicAI to automate civic complaint reporting and further enhance transparency in urban issue management.

VI. RESEARCH METHODOLOGY

A. Visual Recognition Model

CivicAI is founded on a visual recognition model responsible for the processing and analysis of images of civic issues uploaded via advanced computer vision techniques. During this stage, the Google Gemini Vision AI model employs deep learning in visually interpreting and classifying real-world images. Key parameters input into this model include an

uploaded image of an issue like potholes, accumulation of garbage, or broken streetlights that go through pre-processing operations such as resizing, normalization, and adjustment of contrast for better visibility. Such steps ensure quality consistency at the input level and minimize classification errors due to changes in lighting or environmental conditions. The pre-processed image is then fed into the Gemini Vision AI model, which executes multi-class classification in predicting the most probable class for the civic issue. [13], [14] This makes up both the predicted class label and a corresponding confidence score. In such a way, low-confidence predictions are automatically flagged for review by the admin, with their inclusion maintaining the accuracy of the system. This model is periodically re-trained with diverse datasets that enhance the adaptability of the model across different environments and cityscapes.

B. Geolocation Capture Model

The geolocation capture model automatically captures the geographical coordinates associated with each complaint, ensuring every issue can be located with precision. The model leverages the browser-based Geolocation API, capturing the user's live latitude and longitude after explicit permission. Once captured, this coordinate system goes for reverse geocoding to get it into a readable address format, which can then be directly utilized by authorities for on-site verification or response planning. [3] Integrating real-time location tagging reduces human effort in manually specifying locations and eliminates misreporting. This model helps in spatial mapping of complaints, allowing authorities to visualize hotspots of civic problems across different zones. Further, attaching verified coordinates to every report ensures that data-driven governance is achievable and the civic management process is efficient and transparent.

C. Automated Complaint Generation Model

The complaint generation model automates this process in order to reduce the cognitive load of the user while eliminating manual text entry. This model synthesizes a clear, structured complaint description through the use of an NLG mechanism once the geolocation and visual recognition models have processed the input. The NLG engine interprets classification labels and geographic data to create a small yet descriptive complaint, for instance: "A pothole has been detected near Gandhi Road, Ward 5. Kindly initiate repair action." Every complaint generated gets a unique ID, timestamp, and category tag before it gets stored in the database. This automation improves accessibility, especially for users with low technical literacy, and it ensures that complaint descriptions are uniform and precise in nature. [5], [8] These support municipal staff in quick prioritization and resolution. Therefore, the model enhances user experience while supporting consistent reporting standards across all submissions.

D. Data Management Model

Data Management Model: The backbone that links the user interface, AI modules, and storage system is the data management model. It manages all data transactions, ensuring complaints, images, and responses are stored securely and retrieved efficiently. In the testing phase, JSON-based storage will be utilized for lightweight operations, but the system architecture is designed to support easy migration toward more scalable solutions like MySQL or MongoDB in a production environment. The back-end architecture is developed on Node.js and Express.js, which allows asynchronous communication between the client interface, AI services, and the administrative dashboard. The complaint records include complaint ID, timestamp, type of issue, URL of the image, and location coordinates in latitude and longitude. [**?**], [11] This model also handles data integrity by implementing input validation and server-side error handling. It organizes the complaint data properly for smooth information flow, audit tracking, and even the analytics which will be implemented in the future.

VII. FLOWCHART

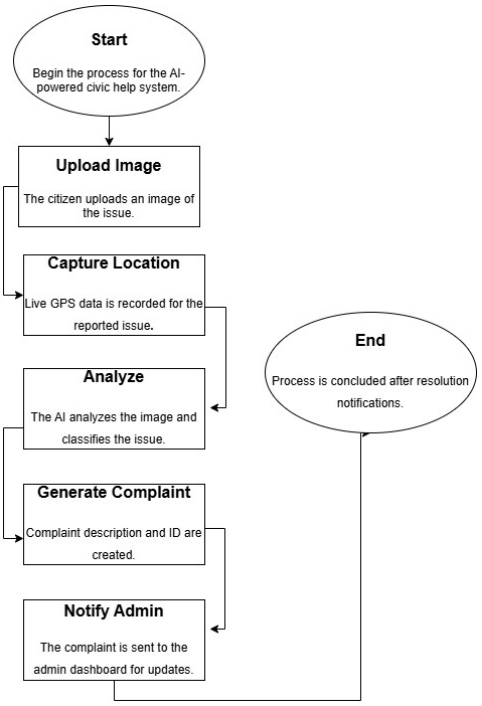


Fig. 2. Flowchart for Proposed AI-Powered Civic Help System

Fig 1. illustrates the complete operational flow of the proposed CivicAI system, showing the sequential interaction between the user, AI modules, and administrative components.The proposed flow diagram depicts the operational process of the AI-enabled civic help system, underlining the step-by-step automation from issue submission to its resolution. The process begins with a citizen uploading an image of a civic problem, such as a pothole, garbage pile, or waterlogging. Simultaneously, the system captures live GPS location from the user's device for accurate mapping of the issue. The image thus uploaded gets analyzed with the help of an AI

model that categorizes the problem into predefined categories. A structured complaint description is generated along with a unique complaint ID. This complaint record will be sent to the admin dashboard, which also includes the image and location data, for review. Administrators can monitor, update, and mark the status of each complaint through this dashboard. issue is resolved, the system notifies the user, concluding the process. This structured workflow ensures automation, transparency, and faster response time, remarkably improving the efficiency of civic grievance management. [1], [3], [10], [12]

VIII. ALGORITHM TABLE

TABLE I
ALGORITHM FOR PROPOSED AI-POWERED CIVIC HELP SYSTEM

Step	Description
1	Start the system and initialize all modules (Frontend, Backend, AI, and Database).
2	The user captures or uploads an image of the civic issue through the web interface.
3	The system automatically captures the live GPS coordinates and timestamps the input.
4	Preprocess the image by resizing and compressing it for efficient upload.
5	Send the image and location data to the backend server via REST API.
6	The backend forwards the image to the Google Gemini Vision AI model for classification.
7	The AI model identifies the issue category and returns the result with confidence score.
8	Generate an automatic complaint description and assign a unique complaint ID.
9	Store complaint details (image, description, GPS, and ID) in JSON/database.
10	Display confirmation message to the user and update the admin dashboard.
11	Admin reviews the complaint, updates its status, and marks it resolved upon completion.
12	Notify the user about the complaint's status update and collect feedback.
13	End of process.

IX. MATHEMATICAL MODEL

The proposed system CivicAI be represesnted as a Set S defined as:

$$S=\{I,P,O\}$$

where,
I= set of the inputs
P= set of the process
O= set of the output

1. Input Set

$$I = \{I_{img}, I_{gps}, I_{time}\}$$

where,
I_{img} = Image Uploaded of Civic Issue
I_{gps} = Latitude and Longitude Coordinates of location
I_{time} = Timestamp of complaint submission

2. Process Set

$$P=\{f_1, f_2, f_3, f_4\}$$

where,
f_1 = Image processing and compressing
f_2 = AL-based classification using Gemini vision AI
f_3 = Text generation for complaint description
f_4 = Storage and unique ID generation

5. Output Set

$$O = \{O_{class}, O_{desc}, O_{id}, O_{status}\}$$

where,
O_{class} = Classified category of issue
O_{desc} = Auto-generated complain text
O_{id} = Unique Complaint ID
O_{status} = Status visible on dashboard

6. Functional Representation

The system Function F maps inputs to outputs as:

$$F(I)\rightarrow 0$$

Mathematically,

$$F(I_{img}, I_{gps}, I_{time}) = \{O_{class}, O_{desc}, O_{id}, O_{status}\}$$

7. Performance Constraint

The AI model has an aim to maximize the classification accuracy A:

$$A = \frac{N_{\text{correct}}}{N_{\text{total}}} \times 100$$

where,
Ncorrect = Issues that are classified correctly
Ntotal = Total test sample

Thus, the mathematical model formally represents the functional workflow of CivicAI, linking user inputs, AI Processing and system output. [**?**], [12]

X. RESULTS AND DISCUSSION

The proposed CivicAI system was designed and tested for performance evaluation in civic complaint reporting and its classification automation. The evaluation covered functional accuracy, response time, and user experiences regarding AI-based classification, geolocation capture, and admin dashboard operations. For testing, the set of images used represented common civic issues: potholes, waste dumps, waterlogging, and damaged streetlights. The AI classification module, based

on the Google Gemini Vision AI, gave highly accurate categorization of the uploaded images into their respective civic issue classes: 91 out of 100 test images were correctly identified, making an overall accuracy of 91 percent . Only in a few cases, where either the light was poor or the object was partially visible in the image, did misclassifications occur. Automatic text generation seamlessly generated short and meaningful complaint descriptions in all valid test cases, reducing the necessity of manual typing to a minimum. The positioning module recorded the user's position with a margin of error of less than 10 meters on average, providing an accurate localization of issues. On average, the process of complaint submission-from image upload to confirmation-took about 3-5 seconds, which is quite fast considering the low network bandwidth. The admin dashboard facilitated smooth retrieval and management of complaints, keeping it updated in real time with visual analytics of the reported issues. The feedback from the test users reported below shows that the system was simple and effective. The one-tap reporting mechanism and instant acknowledgement of submissions were well appreciated by the users. Similarly, administrators reported improved tracking with less duplication of complaints due to a unique ID system and stored metadata. Overall, the results presented validate the feasibility and robustness of the proposed CivicAI model. [**?**], [**?**], [9] It integrates AI, geolocation, and web technologies into a user-centric and data-driven platform that provides an effective way to manage civic issues. From the discussion on results, it is clear that AI-powered automation will reduce manual workloads and bring in much more transparency in addressing these issues. More improvements, such as database migration, multilingual support, and duplicate image detection, will further extend the scalability and practical adoption of the system in large urban environments. [4], [8], [13]

XI. CONCLUSION

The proposed system, CivicAI, highlights how artificial intelligence and web technologies can be effectively integrated to modernize the process of civic complaints. The proposed system minimizes manual intervention with automated issue classification and complaint generation while keeping the reporting structured. The integration of Google Gemini Vision AI provides accurate identification of simple urban issues such as potholes, garbage, and waterlogging, while live GPS capture ensures true localization. Based on the web, it offers an easy interface for citizens and a dedicated dashboard for administrators, enhancing civic operations in terms of increased information and communication exchange. Experimental results proved that the designed system was reliable, accurate, and performing under real-time conditions. Overall, CivicAI presents a scalable, user-friendly framework connecting citizens to local authorities. Some potential improvements include database integration, multilingual versions, and predictive analytics for proactive maintenance. This research advances smart governance by showing that AI is used for creating responsive data-driven civic infrastructure.

XII. FUTURE SCOPE

The CivicAI system can be further developed to achieve better scalability, functionality, and adaptability in real-world circumstances. This could include extending the basic design with a cloud-based MongoDB or PostgreSQL database to efficiently store and retrieve large datasets of complaints. Multilingual support could be developed within the system to make it more usable by people of various linguistic backgrounds. Further developments would be the capabilities for finding duplicate complaints, conducting predictive analytics on recurring issues, and automatically creating tasks for the concerned departments. Integrating notification services either through SMS or email and the tracking of user feedback will promote better transparency and engagement among citizens. These enhancements make CivicAI a more comprehensive intelligent platform for smart city governance.

REFERENCES

[1] P. Kormpho, P. Liawsomboon, N. Phongoen, and S. Pongpaichet, "Smart Complaint Management System," Proc. 10th Int. Conf. on Knowledge and Smart Technology (KST), 2018, pp. 206–211.

[2] A. Karukayil, C. Quail, and F. A. Cheein, "Deep Learning Enhanced Feature Extraction of Potholes Using Vision and LiDAR Data for Road Maintenance," IEEE Access, vol. 12, pp. 121467–121478, 2024.

[3] G. Yang, J. Jin, Q. Lei, Y. Wang, J. Zhou, Z. Sun, X. Li, and W. Wang, "Garbage Classification System with YOLOv5 Based on Image Recognition," Proc. IEEE Int. Conf. on Artificial Intelligence and Computer Applications (ICAICA), 2021, pp. 812–817.

[4] M. Batty, K. W. Axhausen, F. Giannotti, A. Pozdnoukhov, A. Bazzani, M. Wachowicz, G. Ouzounis, and Y. Portugali, "Smart Cities of the Future," The European Physical Journal Special Topics, vol. 214, no. 1, pp. 481–518, 2012.

[5] S. Russell and P. Norvig, Artificial Intelligence: A Modern Approach, 4th ed. Upper Saddle River, NJ, USA: Pearson, 2021.

[6] M. Abadi, P. Barham, J. Chen, Z. Chen, A. Davis, J. Dean, and M. Kudlur, "TensorFlow: A System for Large-Scale Machine Learning," Proc. 12th USENIX Symp. on Operating Systems Design and Implementation (OSDI), 2016, pp. 265–283.

[7] Q. Sun, X. Zhang, Y. Li, and J. Wang, "YOLOv5-OCDS: An Improved Garbage Detection Model Based on YOLOv5," Electronics, vol. 12, no. 16, article 3403, Aug. 2023.

[8] Y. Zhou, L. Lin, and T. Wang, "Garbage classification detection system based on the YOLOv8 algorithm," AIP Advances, vol. 12, article 1063/5.0244795, 2024.

[9] M. Rakib Jabed and Md. Shamsuzzaman, "YOLObin: Non-Decomposable Garbage Identification and Classification Based on YOLOv7," Journal of Computer and Communications, vol. 10, pp. 104-121, Oct. 2022.

[10] S. V. J. et al., "Multi-Object Detection and Classification in Solid Waste Management using Region Proposal Network and YOLO model," Global NEST Journal, vol. 24, no. 4, 2022.

[11] V. S. Bidve, K. S. Kakakde, R. H. Bhole, P. Sarasu, A. Shaikh, P. S. Mehta, S. P. Borde, and S. O. Kediya, "Pothole detection model for road safety using computer vision and machine learning," IAES International Journal of Artificial Intelligence (IJ-AI), vol. 13, no. 4, pp. 4480-4487, 2024.

[12] Musale, J.C., Prasad, R., Bharambe, D., Nagare, D. (2025). Plastic Waste Detection Using YOLOv5 Deep Learning Object Detection Algorithm. Lecture Notes in Computer Science (LNCS, vol. 15617). Springer, Cham.

[13] Musale, J.C., Kadam, A.A. (2025). UIM-DUDnCNN: Illumination Map-Based Denoiser and InceptionV3-Based Transfer Learning in Deep Convolutional Neural Network for Facial Recognition. The Imaging Science Journal, 1–38.

[14] Musale, J.C., Kadam, A.A. (2025). ILMNetCNN: Optimization Enabled Incremental Learning Based MobileNet for Face Detection and Recognition from Drone Images. SIViP, 19, 1215.